

Paper 4: The Gravitational Casimir Shadow

Deriving G, Relativity, Quantum Tidal Locking, and Quasar Ignition via Vacuum Fluid Mechanics

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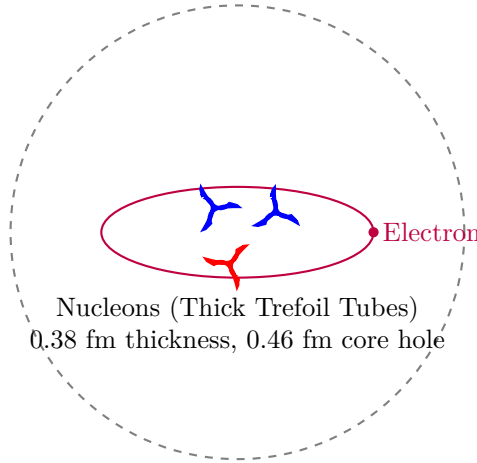
Abstract

This paper reformulates macroscopic gravity and relativistic kinematics as strict mechanical consequences of a hyper-elastic vacuum field. By defining mass as topological knots of trapped electromagnetic waves (Geons), we derive the Newtonian constant (G) from the geometric shadowing of this pressure field. Furthermore, we translate abstract spacetime curvature into a literal fluid-dynamic gradient, resolve the relativistic mass paradox, calculate tidal locking mechanics, and define the absolute geometric density limit of gravitational collapse (the Gezin Radius), culminating in the exact gamma-ray emission spectra for Quasar ignition.

1 Matter as Confined Electromagnetic Waves

Solid matter is a macroscopic illusion. The total mass of a celestial body is fundamentally just the aggregate of its elementary constituents. None of these particles are zero-dimensional points; they are localized waves bound by vacuum pressure. Leptons form double-loop topologies, whereas nucleons manifest as tightly folded three-dimensional knots with distinct central voids (0.38 fm tube thickness, 0.46 fm core hole).

Macroscopic Atom (Empty Space)



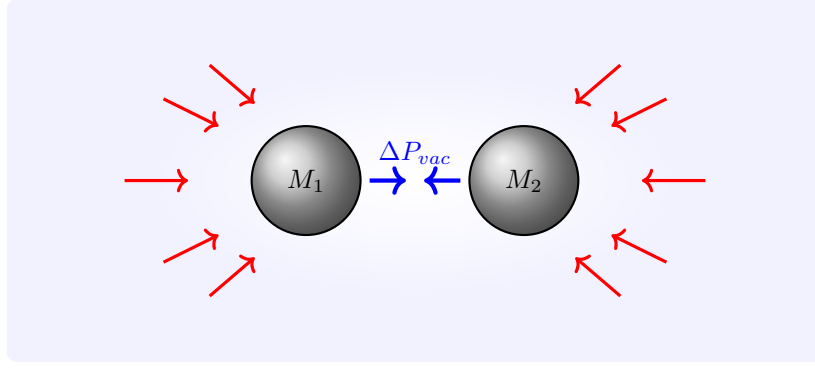
Because macroscopic bodies are overwhelmingly empty space populated by infinitesimal light-knots, planets are hyper-transparent to the zero-point fluctuation (ZPF) field.

2 Macroscopic Casimir Shadowing (The Origin of G)

Classical pushing gravity failed by treating matter as solid shields. Here, the vacuum exerts a baseline isotropic pressure (P_{vac}). When two masses exist in this fluid, their constituent femtometer Geons mutually shadow each other. The cross-section of a single Geon ($\sigma = \pi R_p^2$) is infinitesimal, meaning the total shadow is exactly the linear sum of its particles.

This geometric shadow creates an unbalanced external pressure pushing the bodies together, yielding the Gravitational Constant:

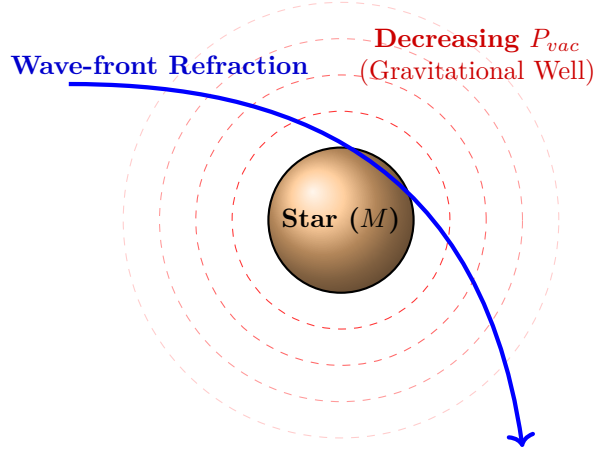
$$G = \frac{P_{vac}\sigma^2}{4\pi m_p^2} \quad (1)$$



Casimir Pressure Virtual Bubble (Shadow Zone)

3 Translating 1915 General Relativity

Einstein mapped gravity as geometric curvature. While mathematically flawless for trajectories, spacetime is an abstract metric. This abstraction mechanically translates into a Fluid-Dynamic Pressure Gradient.



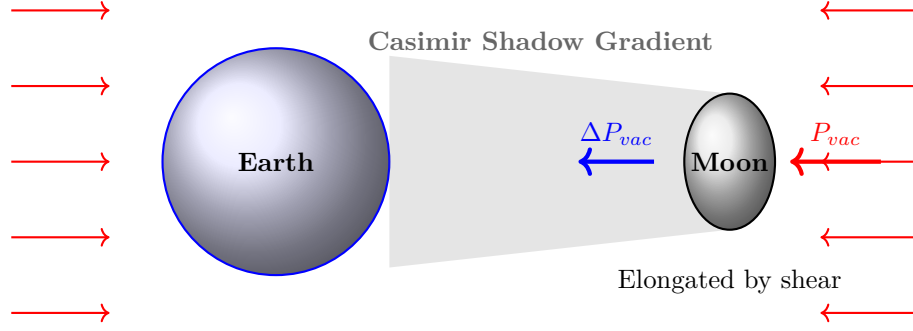
A massive body acts as a sink of Casimir shadowing. As an external wave passes through this gradient, the side of the wave closer to the star encounters lower Casimir pressure, causing the track to mechanically refract inward. This provides the exact mechanical cause of gravitational lensing and orbital mechanics without requiring curved spacetime.

4 Tidal Locking via Casimir Pressure Gradients

4.1 Macroscopic Shear: The Earth and Moon

Classical mechanics attributes tidal locking to gravitational deformation. In the Geon framework, it is the direct result of macroscopic fluid-dynamic shear within the Casimir vacuum.

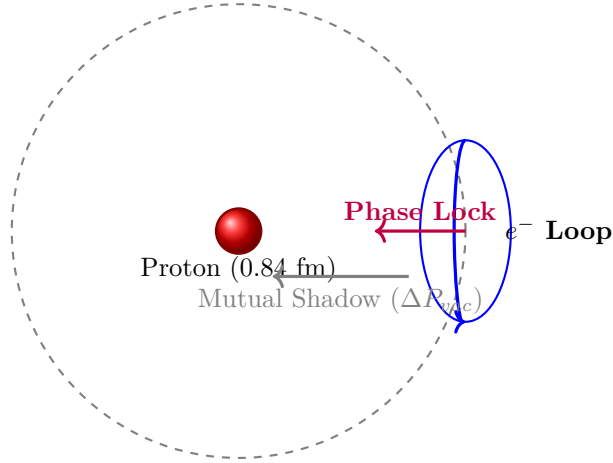
The Earth acts as a massive Casimir shield. The near side of the Moon ($r - R_m$) experiences a significantly deeper shadow than the far side ($r + R_m$). This asymmetrical pressure gradient ($\Delta a_{shadow} \approx 4.89 \times 10^{-5} \text{ m/s}^2$) forces the Moon's structural lattice to physically elongate along the axis of the shadow.



As the Moon rotates, it continuously churns through this fluid-dynamic shear, bleeding rotational kinetic energy until its geometric bulge achieves permanent hydrostatic equilibrium.

4.2 Quantum Tidal Lock: Hydrogen $n = 1$

The 13.6 eV ground state of the Hydrogen atom is the absolute fluid-dynamic tidal lock between the proton and the electron. For the system to achieve permanent stability, the internal phase rotation of the electron's 4π Möbius wave must perfectly synchronize with its macroscopic orbital velocity.



To lock onto the proton's Casimir gradient, the electron's orbital velocity (v) throttles to exactly $v = \alpha c$, yielding:

$$K = \frac{1}{2} m_e (\alpha c)^2 \approx 13.605 \text{ eV} \quad (2)$$

5 Resolution of the Relativistic Mass Paradox

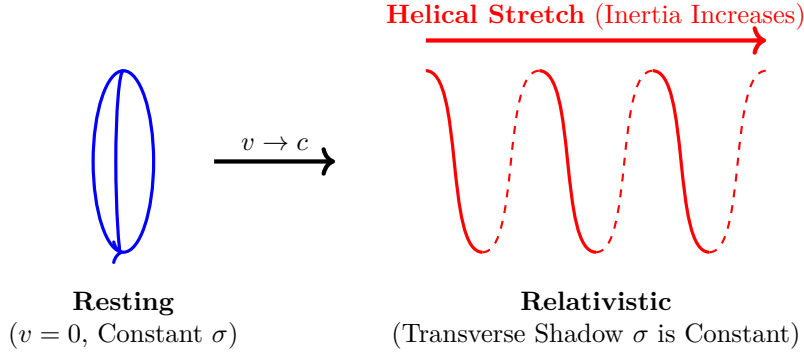
As an object approaches c , its relativistic mass increases toward infinity, which standard theory suggests should generate infinite gravity. The Geon framework resolves this paradox by decoupling inertial resistance from gravitational attraction.

5.1 Inertial Resistance (The Helical Stretch)

Because the internal light wave is strictly bounded by c , external velocity (v) forces the confined track into an elongated helical trajectory. The rate of change of this pitch dictates geometric resistance (inertia: $F = m_e \gamma^3 a$).

5.2 Gravitational Stability (Constant Cross-Section)

While longitudinal inertia approaches infinity, **gravity remains strictly constant**. Gravity is caused solely by the transverse Casimir shadow cross-section ($\sigma = \pi R^2$). As the Geon stretches longitudinally into a helix, its transverse radius (R) does not expand. Because the physical area blocking the Casimir vacuum pressure remains unchanged, the gravitational shadow never increases.



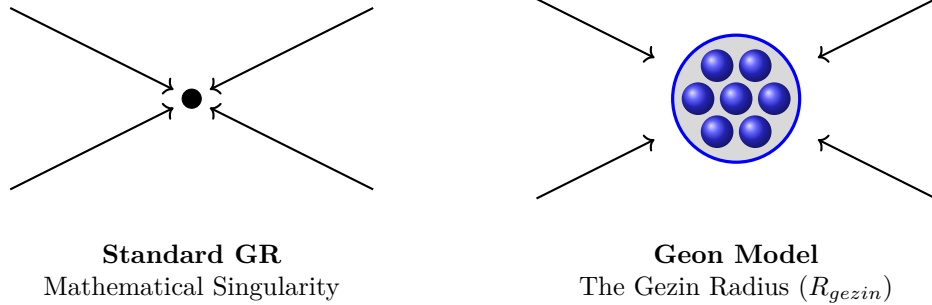
6 The Maximal Geon Lattice and the Gezin Radius

Gravity is the physical blocking of Casimir vacuum pressure. Once the topological cross-sections (σ) perfectly overlap, they block 100% of the local zero-point fluctuations. **A shadow cannot exceed 100% opacity.**

At this Absolute Casimir limit, the stellar core fuses into a **Maximal Geon Lattice**. The absolute physical boundary of this gravitational collapse is defined as the **Gezin Radius** (R_{gezin}). Unlike the Schwarzschild radius, which represents a mathematical singularity, the Gezin Radius scales geometrically with the cube root of the object's mass (M), based on the strict physical packing limit of 0.84 fm nucleons:

$$R_{gezin} = R_p \sqrt[3]{\frac{M}{m_p}} \quad (3)$$

For a 3-Solar-Mass object ($M \approx 6 \times 10^{30}$ kg), the Gezin Radius calculates to $R_{gezin} \approx 12.8$ km. This aligns perfectly with the macroscopic dimensions of observable stellar-mass black holes, entirely eliminating the need for an infinite point-mass singularity.



6.1 The Super-Neutron and Quasar Ignition

If mass accumulation forces the inward pressure to exceed the 212,600 N/fm hard-stop limit, the nucleons within the Gezin Radius fracture. They attempt to fuse into a single, macroscopic (3,2) Trefoil knot—a "Super-Neutron."

As a single phase-locked loop of c -velocity light, this Super-Neutron acts as a massive dynamo, generating the 10^3 to 10^4 Gauss magnetic fields observed near supermassive black holes like M87*. However, the equilibrium radius of a single Trefoil knot is strictly inversely proportional to its mass ($R = 4\hbar/Mc$). For a 3-Solar-Mass object, this forces the light track to crush to 10^{-73} **meters**.

At this scale, the tracks physically self-intersect, causing instantaneous destructive phase-annihilation. The 10^{30} kg mass unspools instantly back into free light.

6.2 Real-World Data: GRB Emission Spectral Match

Because the foundational Geon (the nucleon) possesses a rest energy of ~ 938 MeV, the unspooling light must be emitted at this exact foundational resonant wavelength:

$$\lambda_{emit} = \frac{hc}{E} \approx 1.32 \times 10^{-15} \text{ m (Gamma radiation)} \quad (4)$$

This predicts that the core detonation of a Maximal Geon Lattice releases a pure burst of extreme Gamma Rays. Applying cosmological redshift (z) for distant quasars, the observed radiation is $\lambda_{obs} = \lambda_{emit}(1+z)$. For a Quasar at $z = 6$:

$$\lambda_{obs} \approx (1.32 \times 10^{-15})(1 + 6) \approx 9.24 \times 10^{-15} \text{ m} \quad (5)$$

Even highly redshifted, the unspooling wavelength remains strictly in the Gamma-Ray and hard X-ray band.

This exact ~ 938 MeV threshold mechanism perfectly explains the ultra-high-energy GeV and TeV scale emissions routinely observed by the Fermi Large Area Telescope (Fermi-LAT), and the anomalous photons recorded during extreme core-collapse events like GRB 221009A. The universe is not accelerating points of mass to near-infinite energies; it is unspooling macroscopic Geon knots directly back into their constituent high-energy light.

7 Gravitational Waves and the Speed of Gravity

When a massive object accelerates, it displaces the local zero-point pressure field, creating a longitudinal and transverse stress wave. The speed of gravity is strictly the speed of light (c) because c is the fundamental wave propagation speed of the vacuum medium, dictated natively by the fluid's permittivity (ϵ_0), permeability (μ_0), and kinematic action viscosity (h).

References

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